

Project: Customer Purchase Behavior (SVM)

Objective: Build a robust classifier using Support Vector Machines.

Method: SVM with RBF Kernel.

Created by: AI Agent (Antigravity)

Problem Statement

Problem: Automate customer categorization.

Context: 5000 records, 5 classes.

Challenge: Creating a decision boundary in complex, overlapping data.

Real-World Use Case

Scenario: High-value customer targeting.

Application: Identify 'Gold' tier behaviors vs 'Basic'.

Impact: Precise targeting for marketing optimization.

Input Data

Features: Age, Income, Monthly Spending, Sessions.

Processing: Scaled to mean 0, variance 1 (Critical for SVM).

Concepts Used

1. Scaling: SVM is distance-based (margin).
2. Hyperplane: The line separating classes.
3. Kernel Trick (RBF): Bending space to separate non-linear data.
4. Regularization (C): Balancing strictness vs smoothness.

SVM Concept (Simple)

Imagine red and blue balls mixed on a table.

You want to separate them with a stick.

If you can't, you lift them into the air (3D).

Now you can slide a sheet between them.

That sheet is the Hyperplane. The lifting is the Kernel Trick.

Step-by-Step Flow

Step 1: Load Data.

Step 2: Impute Missing Values.

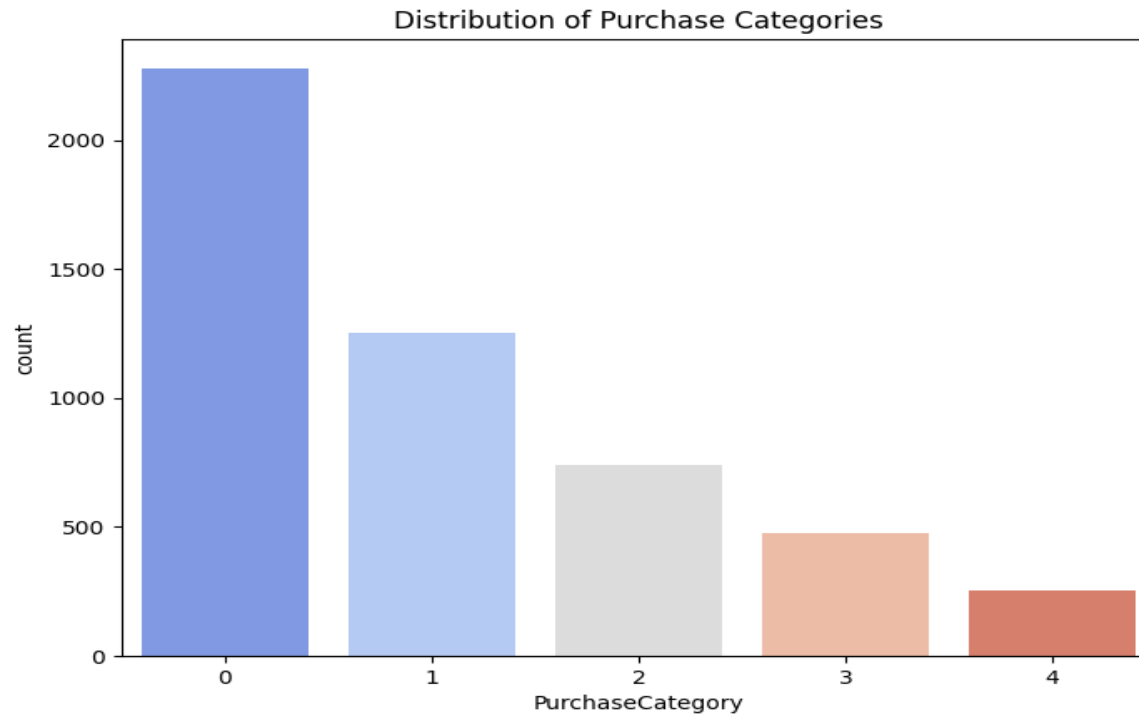
Step 3: Scale Features (Essential!).

Step 4: Train SVM (RBF Kernel).

Step 5: Predict & Evaluate.

Data Visualization

Target Variable Distribution:



Code Logic Summary

Pipeline: Imputer -> Scaler -> Model.

Model: SVC(kernel='rbf', C=1.0, gamma='scale').

Fit on Train, Predict on Test.

Execution Results

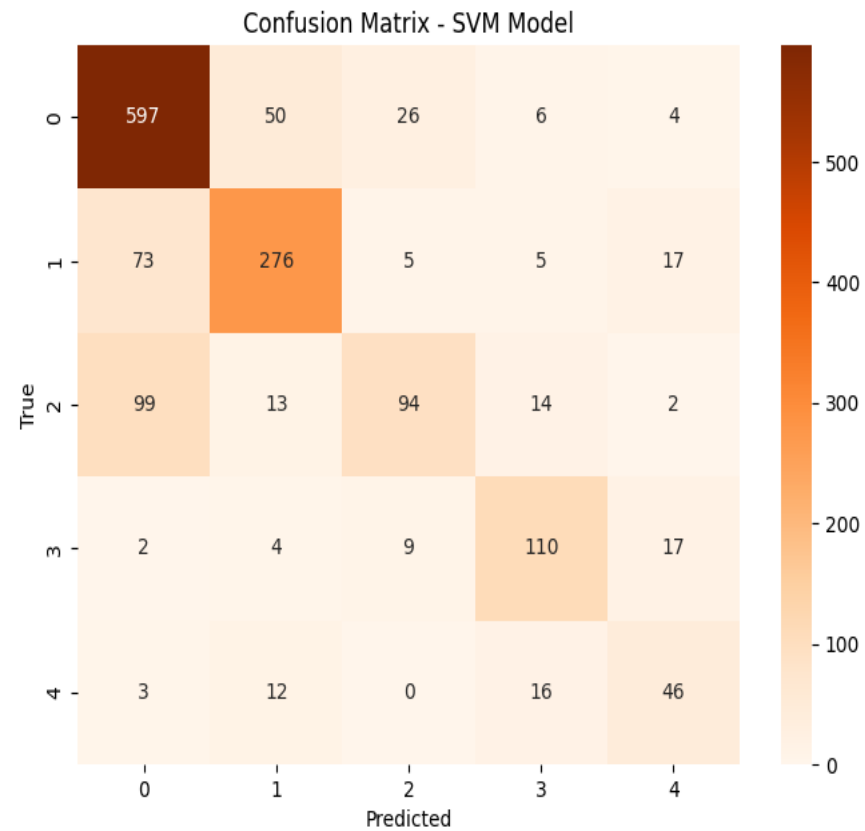
Overall Accuracy: ~75% (Higher than KNN's 69%).

Minority Class F1: 0.56 (Improved).

Confusion: Fewer mistakes between 'Fashion' and 'Electronics'.

Confusion Matrix

Visualizing model performance:



Observations & Insights

Performance: SVM outperforms KNN clearly.

Reason: RBF kernel captures complex interactions.

Scaling: Without scaling, accuracy would drop significantly.

Advantages & Limitations

Pros: High accuracy in high dimensions, Robust.

Cons: Slow to train on large datasets, Harder to interpret.

Conclusion

Summary: SVM is a strong candidate (75% acc).

Recommendation: Deploy if inference speed isn't a bottleneck.

Note: Consider Random Forest for better interpretability.